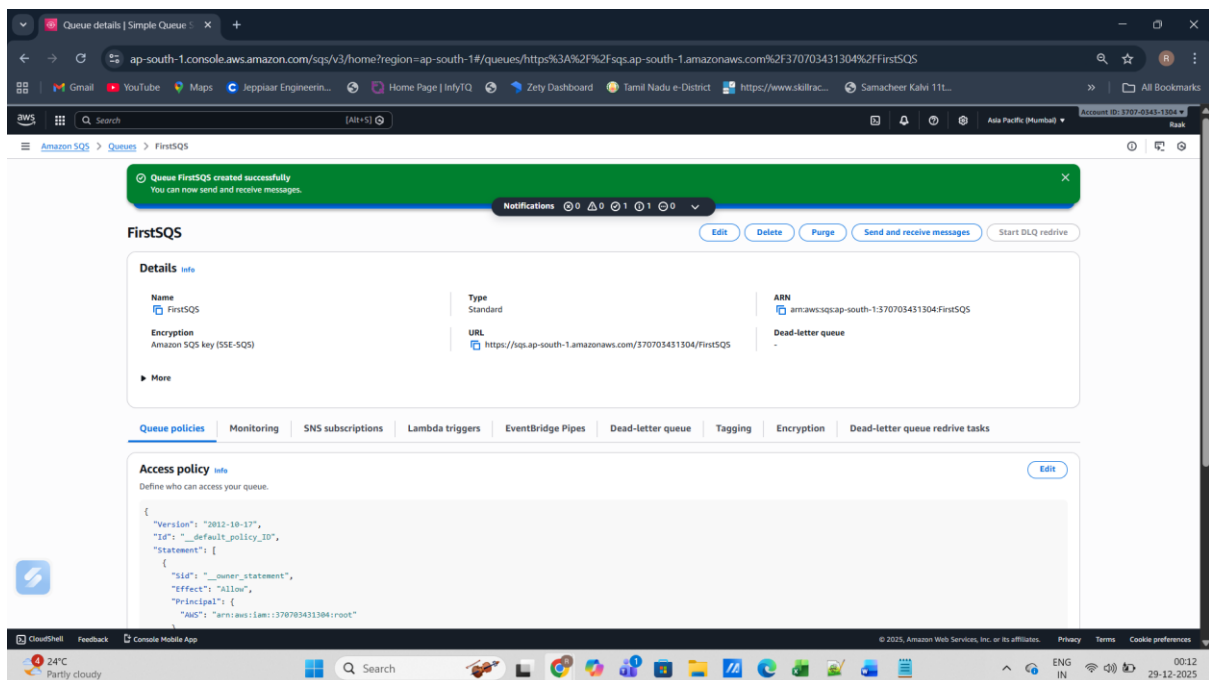


WEEK 7 ASSIGNMENT: Serverless Workflow Automation Project

Objective:

The objective of this project is to design and implement an event-driven serverless architecture using AWS services. The system processes messages sent to an SQS queue using AWS Lambda, sends email notifications via SNS upon successful execution, and monitors errors using CloudWatch alarms.

SQS created:



SNS configured:

The screenshot shows the Amazon SNS console in a web browser. The URL is `ap-south-1.console.aws.amazon.com/sns/v3/home?region=ap-south-1#/subscription/arn:aws:sns:ap-south-1:370703431304:FirstSNS:77b2e4b0-6299-484d-b654-418026ef03b8`. The page displays a confirmation message: "Subscription to FirstSNS created successfully. The ARN of the subscription is arn:aws:sns:ap-south-1:370703431304:FirstSNS:77b2e4b0-6299-484d-b654-418026ef03b8." Below this, the subscription details are shown:

Details	
ARN arn:aws:sns:ap-south-1:370703431304:FirstSNS:77b2e4b0-6299-484d-b654-418026ef03b8	Status Pending confirmation
Endpoint raaksh2405@gmail.com	Protocol EMAIL
Topic FirstSNS	
Subscription Principal arn:aws:iam::370703431304:root	

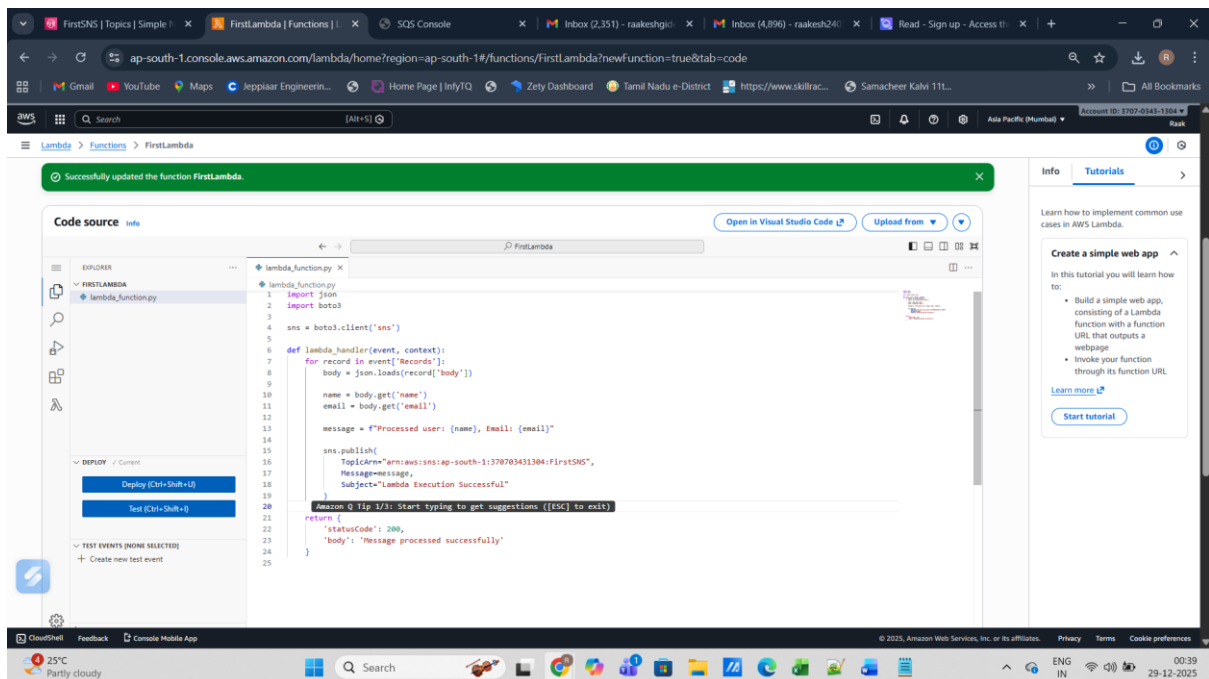
Below the details, there are tabs for "Subscription filter policy" and "Redrive policy (dead-letter queue)". The "Subscription filter policy" tab is active, showing a message: "No filter policy configured for this subscription. To apply a filter policy, edit this subscription." The left sidebar shows the Amazon SNS navigation menu, including Dashboard, Topics, Subscriptions, and Mobile. The bottom of the browser shows the Windows taskbar with the date 29-12-2025 and time 00:24.

The screenshot shows an email from AWS titled "Subscription confirmed!". The email content is as follows:

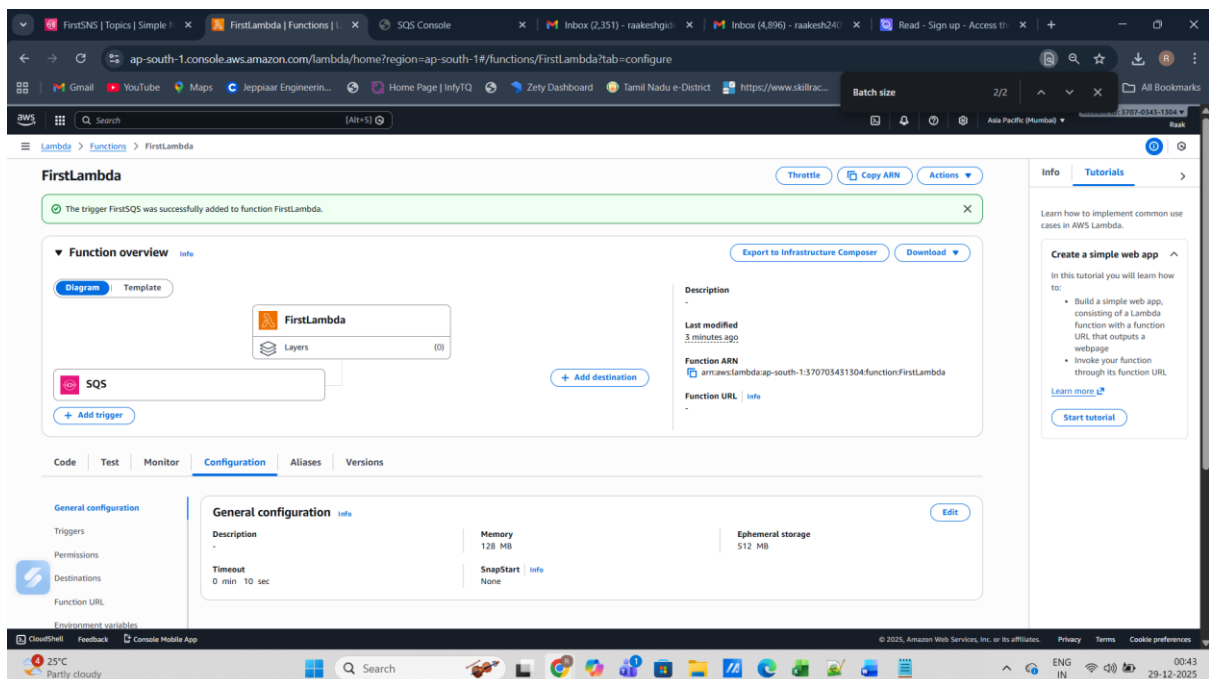
Subscription confirmed!
You have successfully subscribed.
Your subscription's id is:
arn:aws:sns:ap-south-1:370703431304:FirstSNS:f054639e-e6f9-44c1-9c95-bca8f1ac875d
If it was not your intention to subscribe, [click here to unsubscribe](#).

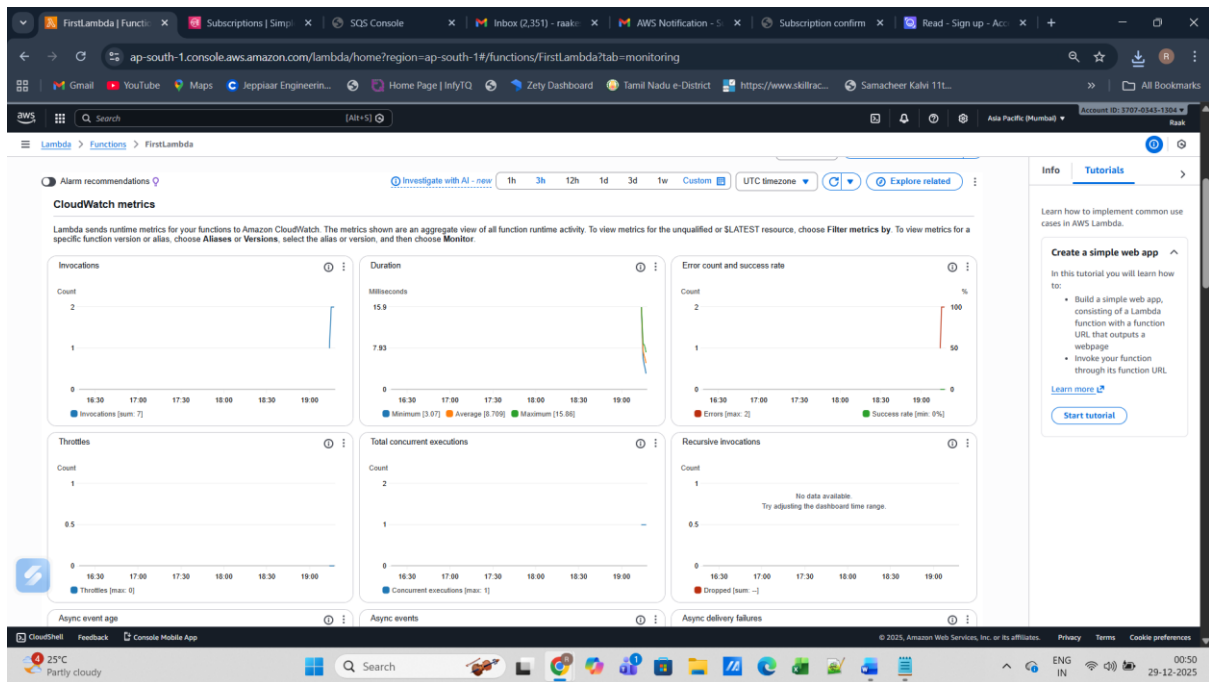
The email is displayed in a web browser window. The browser's address bar shows the URL: `sns.ap-south-1.amazonaws.com/confirmation.html?topicArn=arn:aws:sns:ap-south-1:370703431304:FirstSNS&token=2336412f37fb687f5d51e6e2425a8a5973989d56678250c61587827c1f49...`. The left sidebar of the browser shows the AWS logo and the text "Simple Notification Service". The bottom of the browser shows the Windows taskbar with the date 29-12-2025 and time 00:49.

Lambda configured:

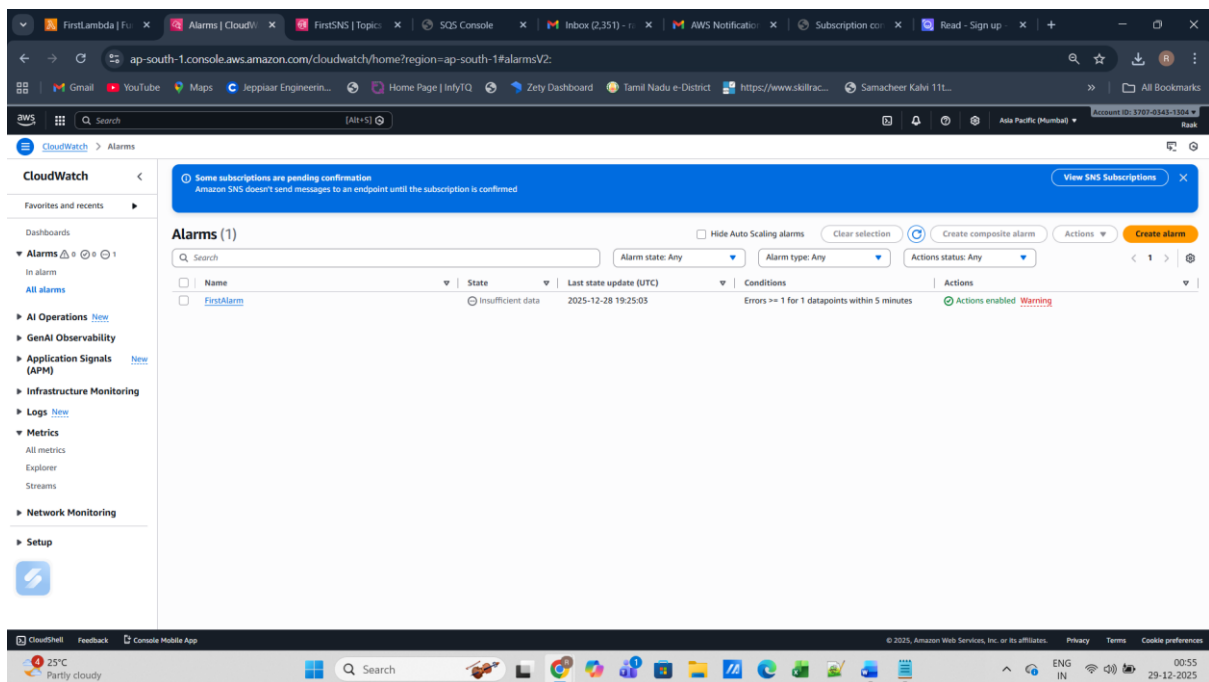


SQS triggered through lambda:





Alarm created in association with cloudwatch:



Cloud formation YAM template:

AWSTemplateFormatVersion: '2010-09-09'

Description: Week 7 - Lambda with SQS and SNS

Resources:

MySQSQueue:

Type: AWS::SQS::Queue

MySNSTopic:

Type: AWS::SNS::Topic

MyLambdaRole:

Type: AWS::IAM::Role

Properties:

AssumeRolePolicyDocument:

Version: '2012-10-17'

Statement:

- Effect: Allow

Principal:

Service: lambda.amazonaws.com

Action: sts:AssumeRole

ManagedPolicyArns:

- arn:aws:iam::aws:policy/AmazonSQSFullAccess
- arn:aws:iam::aws:policy/AmazonSNSFullAccess
- arn:aws:iam::aws:policy/CloudWatchLogsFullAccess

MyLambdaFunction:

Type: AWS::Lambda::Function

Properties:

Runtime: python3.9

Handler: index.lambda_handler

Role: !GetAtt MyLambdaRole.Arn

Code:

ZipFile: |

```
def lambda_handler(event, context):
```

```
    print("Lambda triggered from SQS")
```

```
    print("Lambda triggered from SQS")
```